STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-CARTESIAN-FUNCTORIAL

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6204-6212 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 6214-6263, the complete author proof (48 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 12 of queue_sdga.json (pb=2053, stmt 6204-6212, proof 6214-6263).

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-cartesian-functorial' also exists in stacks-sites-cohomology — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0FTI: sources/stacks-sdga/pinned-original-source.tex, source lines 4896–4910. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $g : (Sh(\mathcal{C}'), \mathcal{O}') \to (Sh(\mathcal{C}), \mathcal{O})$ and $\varphi : g^*\mathcal{A} \to \mathcal{A}'$ be as above. Then the functor $Lg^* : D(\mathcal{A}, d) \to D(\mathcal{A}', d)$ maps $QC(\mathcal{A}, d)$ into $QC(\mathcal{A}', d)$.*

Proof. Let $U' \in \mathrm{Ob}(\mathcal{C}')$ with image $U = u(U')$ in $\mathcal{C}$. Let pt denote the category with a single object and a single morphism. Denote $(Sh(pt), \mathcal{O}'(U'))$ and $(Sh(pt), \mathcal{O}(U))$ the ringed topoi as indicated endowed with the differential graded algebras $\mathcal{A}'(U)$ and $\mathcal{A}(U)$. Of course we identify the derived category of differential graded modules on these with $D(\mathcal{A}'(U'), \mathrm{d})$ and $D(\mathcal{A}(U), \mathrm{d})$. Then we have a commutative diagram of ringed topoi

$$\begin{array}{ccc} (Sh(pt), \mathcal{O}'(U')) & \xrightarrow{U'} & (Sh(\mathcal{C}'), \mathcal{O}') \\ \downarrow & & \downarrow g \\ (Sh(pt), \mathcal{O}(U)) & \xrightarrow{U} & (Sh(\mathcal{C}), \mathcal{O}) \end{array}$$

each endowed with corresponding differential graded algebras. Pullback along the lower horizontal morphism sends M in $D(\mathcal{A}, \mathrm{d})$ to $R\Gamma(U, K)$ viewed as an object in $D(\mathcal{A}(U), \mathrm{d})$. Pullback by the left vertical arrow sends M to $M \otimes^{\mathbf{L}}_{\mathcal{A}(U)} \mathcal{A}'(U')$. Going

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around the diagram either direction produces the same result (Lemma 0FTI) and hence we conclude

$$R\Gamma(U', Lg^*K) = R\Gamma(U, K) \otimes^{\mathbf{L}}_{\mathcal{A}(U)} \mathcal{A}'(U')$$

Finally, let $f' : U' \to V'$ be a morphism in $\mathcal{C}'$ and denote $f = u(f') : U = u(U') \to V = u(V')$ the image in $\mathcal{C}$. If K is in $QC(\mathcal{A}, \mathrm{d})$ then we have

$$\begin{aligned} R\Gamma(V', Lg^*K) \otimes^{\mathbf{L}}_{\mathcal{A}'(V')} \mathcal{A}'(U') &= R\Gamma(V, K) \otimes^{\mathbf{L}}_{\mathcal{A}(V)} \mathcal{A}'(V') \otimes^{\mathbf{L}}_{\mathcal{A}'(V')} \mathcal{A}'(U') \\ &= R\Gamma(V, K) \otimes^{\mathbf{L}}_{\mathcal{A}(V)} \mathcal{A}'(U') \\ &= R\Gamma(V, K) \otimes^{\mathbf{L}}_{\mathcal{A}(V)} \mathcal{A}(U) \otimes^{\mathbf{L}}_{\mathcal{A}(U)} \mathcal{A}'(U') \\ &= R\Gamma(U, K) \otimes^{\mathbf{L}}_{\mathcal{A}(U)} \mathcal{A}'(U') \\ &= R\Gamma(U', Lg^*K) \end{aligned}$$

as desired. Here we have used the observation above both for U' and V'. □